

Erdős Problem #161

EP-161	L1 - Tractable	Partiellement résolu	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

Let $\alpha \in [0, 1/2]$ and $n, t \geq 1$. Let $F^{(t)}(n, \alpha)$ be the smallest m such that we can 2 -colour the edges of the complete t -uniform hypergraph on n vertices such that if $X \subseteq [n]$ with $|X| \geq m$ then there are at least $\alpha \binom{|X|}{t}$ many t -subsets of X of each colour.

For fixed n, t as we change α from 0 to $1/2$ does $F^{(t)}(n, \alpha)$ increase continuously or are there jumps? Only one jump?

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